- +SRA- FastQ conversion- each fastq sequence in a separate zip file
- Metadata preparation (sample id, type of sample like healthy or control, gender if present, and any point which is different in the samples)
- Manifest preparation(sample_ID,forward absolute_path,reverse absolute_path){in paired end sequencing)but for single end it s normal

Log for QIIME-SILVA Pipeline

Check SRA Toolkit

Prefetch -- version
fasterq-dump -- version

Download SRA File

prefetch SRRcode (For all samples)

Convert SRA to FASTQ

fasterq-dump SRRcode.sra

Run FASTQC

- If single end — fastqc SRRcode.fastq
- If paired end —-- fastqc SRRcode(1) SRRcode(2) {no brackets or 1,2 in code it is just for reference}

Zip Fastq

gzip SRRcode.fastq (for all fastq files it helps to sustain storage)

Importing fastq sequences

- qiime tools import \

> --type 'SampleData[PairedEndSequencesWithQuality]' \
> --input-path manifest1.tsv \
> --output-path demux.qza \
> --input-format PairedEndFastqManifestPhred33V2

Using PEAR(if needed only otherwise skip)

pear -f SRRXXXXXXX_1.fastq \
-r SRRXXXXXXX_2.fastq \
-o SRRXXXXXXX

Summerizing Demultiplex results

- qiime demux summarize \
  > --i-data demux.qza \
  > --o-visualization demux.qza

Visualization of demux

- qiime demux summarize \
  --i-data demux.qza \
  --o-visualization demux.qzv

For visualizing any file – `qiime tools view ______.qzv`

Denoising by DADA2

qiime dada2 denoise-paired \
--i-demultiplexed-seqs demux.qza \
--p-trim-left-f 0 \
--p-trim-left-r 0 \
--p-trunc-len-f 280 \
--p-trunc-len-r 260 \
--o-table table1-dada2.qza \
--o-representative-sequences rep-seqs-dada21.qza \
--o-denoising-stats denoising-stats1.qza

Metadata of denoising

qiime metadata tabulate \
> --m-input-file denoising-statsfinal.qza \
> --o-visualization denoising-statsfinal.qzv

- **Then visualizing qzv**

Then moving files to original file for feature table

mv rep-seqs-dada2.qza rep-seqs.qza
mv table-dada2.qza table.qza

Feature table summaries(according to the file names)

qiime feature-table summarize \
  --i-table table.qza \
  --o-visualization table.qzv \
  --m-sample-metadata-file sample-metadata.tsv
        qiime feature-table tabulate-seqs \
  --i-data rep-seqs.qza \
  --o-visualization rep-seqs.qzv

Generate a tree for phylogenetic diversity analyse

qiime phylogeny align-to-tree-mafft-fasttree \
  --i-sequences rep-seqs.qza \
  --o-alignment aligned-rep-seqs.qza \
  --o-masked-alignment masked-aligned-rep-seqs.qza \
  --o-tree unrooted-tree.qza \
  --o-rooted-tree rooted-tree.qza

Tree visualisation

qiime tools export \ (QIIME exports rooted phylogenies as **Newick format**.)
--input-path rooted-tree.qza \
--output-path exported-tree

Then go to itol and put tree.nwk in it and visualize it

ALPHA & BETA diversity analysis

➔ **ALPHA diversity metrics**

- **Phylogenetic Diversity**

```
qiime diversity core-metrics-phylogenetic \
  --i-phylogeny rooted-tree.qza \
  --i-table table.qza \
  --p-sampling-depth 1103 \(can be edited by table.qzv file according to the requirement)
  --m-metadata-file sample-metadata.tsv \(original metadata filtered one)
  --output-dir core-metrics-results
```

- **Faith Phylogenetic Diversity and Evenness(original metadata.tsv is used)**

```
qiime diversity alpha-group-significance \
  --i-alpha-diversity core-metrics-results/faith_pd_vector.qza \
  --m-metadata-file sample-metadata.tsv \
  --o-visualization core-metrics-results/faith-pd-group-significance.qzv

qiime diversity alpha-group-significance \
  --i-alpha-diversity core-metrics-results/evenness_vector.qza \
  --m-metadata-file sample-metadata.tsv \
  --o-visualization core-metrics-results/evenness-group-significance.qzv
```

➔ **Beta diversity metrics**

- **Unweighted UniFrac Site Significance(in this code one column is used if have more columns then again the same code but in the place of Disease the column name will come**)

```
qiime diversity beta-group-significance \
> --i-distance-matrix core-metrix-results/unweighted_unifrac_distance_matrix.qza \
> --m-metadata-file matadata.tsv \
> --m-metadata-column Disease \
>  --o-visualization core-metrix-results/unweighted-unifrac-significance.qzv \
> --p-pairwise
```

- **Emporer Plots**

```
qiime emperor plot \
-i-pcoa core-metrics-results/unweighted_unifrac_pcoa_results.qza \
--m-metadata-file sample-metadata.tsv \
--p-custom-axes days-since-experiment-start \
```

--o-visualization core-metrics-results/unweighted-unifrac-emperor-days-since-experiment-start.qzv

qiime emperor plot \
--i-pcoa core-metrics-results/bray_curtis_pcoa_results.qza \
--m-metadata-file sample-metadata.tsv \
--p-custom-axes days-since-experiment-start \ -
-o-visualization core-metrics-results/bray-curtis-emperor-days-since-experiment-start.qzv

Visualization of pcoa plots (same code for all plots only names changes)

- **Data export**

  qiime tools export \
  --input-path (pcoa_results.qza (file)) \ (no brackets it is just for understanding)
  --output-path unweighted_pcoa_export

- **Loading PCoA coordinates in R for plotting**

  pcoa <- read.table({we are storing results in a variable pcoa }
  "ordination.txt",
  skip = 9,
  header = FALSE,(because of no sample id line)
  row.names = 1,
  fill = TRUE(if any column have incomplete values then it will complete it)
)

- **dim(pcoa) (it shows 64 rows and column)**
- **head(pcoa)(it will show te starting coordinated data of pcoa)**
- **rownames(pcoa)(it shows all sample ids)**
- **Load Metadata**
  metadata <- read.table(
  "metadata.tsv",
  header = TRUE,
  sep = "\t",
  row.names = 1
)

- **head(metadata)**

- **Merge metadata and pcoa coordinates**
  data <- merge(pcoa, metadata, by = "row.names")
- **Check Merged Data**

- **head(data)**

- **Library call**
  library(ggplot2)
- **Plotting with group colouring**
  ggplot(data, aes(x = V2, y = V3, color = Disease,shape = Disease)) + { +
  tells add on)
  geom_point(size = 4) + {size = size of dot }
  theme_classic() +
  labs(
  title = "Bray-Curtis PCoA",
  x = "PC1",
  y = "PC2"
)

  (In plotting the code is same for every plot but only specific files and file names differ)

- **Alpha Rarefraction Plot**

```
qiime diversity alpha-rarefaction \
  --i-table table.qza \
  --i-phylogeny rooted-tree.qza \
  --p-max-depth 4000 \ (It depends on table.qzv file)
  --m-metadata-file sample-metadata.tsv \
  --o-visualization alpha-rarefaction.qzv
```

Taxanomy Analysis

- **Downloading SILVA Database**

wget https://data.qiime2.org/2023.9/common/silva-138-99-tax.qz

wget https://data.qiime2.org/2023.9/common/silva-138-99-tax.qza

(OR)
Download it from Silva website with respective to qiime version compatibility.

(OR)
Download Silva.fasta file (it should be ssu because it is for 16s)

- **Convert Silva RNA to DNA**
  sed 's/U/T/g' SILVA_138.2_SSURef_NR99_tax_silva_trunc.fasta > silva_dna.fasta

- **Import sequence into qiime**

```
qiime tools import \
--type 'FeatureData[Sequence]' \
--input-path silva_dna.fasta \
--output-path silva-seqs.qza
```

- **Extract taxonomy from FASTA headers**
  grep ">" silva_dna.fasta | sed 's/>//' | awk -F" " '{print $1"\t"$2}' > silva-taxonomy.tsv

- **Import Taxonomy**

```
qiime tools import \
--type 'FeatureData[Taxonomy]' \
--input-format HeaderlessTSVTaxonomyFormat \
--input-path silva-taxonomy.tsv \
--output-path silva-taxonomy.qza
```

- **Extract V4 region(because in every papers v4 is referred)**

```
qiime feature-classifier extract-reads \
--i-sequences silva-seqs.qza \
--p-f-primer GTGCCAGCMGCCGCGGTAA \
--p-r-primer GGACTACHVGGGTWTCTAAT \
--p-trunc-len 250 \
--o-reads silva-v4-seqs.qza
```

- **Train Classifier**

```
qiime feature-classifier fit-classifier-naive-bayes \
> --i-reference-reads silva-v4-seqs.qza \
> --i-reference-taxonomy silva-taxonomy.qza \
> --o-classifier silva-classifier.qza
```

- **Assign classification taxonomy**

```
qiime feature-classifier classify-sklearn \
> --i-classifier silva-classifier.qza \
> --i-reads rep-seqs-dada21.qza \
> --o-classification taxonomy.qza
```

- **Visualization of taxonomy**
  qiime metadata tabulate \

> --m-input-file taxonomy.qza \
> --o-visualization taxonomy.qzv

- **Taxonomy Barplot**
  qiime taxa barplot \
  > --i-table table1-dada2.qza\
  > --i-taxonomy taxonomy.qza \
  > --m-metadata-file matadata.tsv \
  > --o-visualization taxa-bar-plots.qzv

    - **View the plot**

Taxonomic Table
qiime tools export \
  --input-path taxonomy.qza \
  --output-path exported-taxonomy

Read Count Table
qiime tools export \
--input-path table.qza \
--output-path exported-table (in this file feature-table.biom this should be present)

–Converting biom to tsv:-
biom convert \
-i exported-table/feature-table.biom \
-o feature-table.tsv \
--to-tsv

Collaqpse at genus level
qiime taxa collapse \
--i-table table.qza \
--i-taxonomy taxonomy.qza \
--p-level 6 \
--o-collapsed-table genus-table.qza

–Export genus table

qiime tools export \
--input-path genus-table.qza \
--output-path genus-export

–Convert to tsv:-

biom convert \
-i genus-export/feature-table.biom \
-o genus-table.tsv \

--to-tsv

Collapse at phylum level

```
qiime taxa collapse \
--i-table table.qza \
--i-taxonomy taxonomy.qza \
--p-level 2 \
--o-collapsed-table phylum-table.qza

qiime tools export \
--input-path phylum-table.qza \
--output-path phylum-export

biom convert \
-i phylum-export/feature-table.biom \
-o phylum-table.tsv \
--to-tsv
```

Filter only bacteria

```
qiime taxa filter-table \
--i-table table.qza \
--i-taxonomy taxonomy1.qza \
--p-exclude Eukaryota,Animalia,Metazoa,mitochondria,chloroplast \
--o-filtered-table bacteria-asv-table.qza

qiime taxa collapse \
--i-table bacteria-asv-table.qza \
--i-taxonomy taxonomy.qza \
--p-level 6 \
--o-collapsed-table bacteria-genus-table.qza

qiime feature-table relative-frequency \
--i-table bacteria-genus-table.qza \
--o-relative-frequency-table relative-bacteria-genus.qza

qiime tools export \
--input-path relative-bacteria-genus.qza \
--output-path rel-genus-export

biom convert \
```

-i rel-genus-export/feature-table.biom \
-o relative-bacteria-genus.tsv \
--to-tsv

Disease-only core

qiime feature-table core-features \
--i-table bacteria-genus-table.qza \
--p-min-fraction 0.7 \
--p-max-fraction 1.0 \
--o-visualization disease-core.qzv

Differential abundance

qiime composition add-pseudocount \
--i-table bacteria-genus-table.qza \ (for pseudocount)
--o-composition-table comp-genus.qza

qiime composition ancom \
--i-table comp-genus.qza \
--m-metadata-file metadata.tsv \
--m-metadata-column Group(any group in metadata we want to observe) \
--o-visualization ancom-disease-vs-control.qzv

PERMANOVA significance

qiime diversity beta-group-significance \
--i-distance-matrix core-metrics-results/bray_curtis_distance_matrix.qza \
--m-metadata-file metadata.tsv \
--m-metadata-column disease \
--p-method permanova \
--p-pairwise \
--p-permutations 999 \
--o-visualization bray-permanova-survival.qzv

Machine learning biomarkers

qiime sample-classifier classify-samples \
--i-table bacteria-genus-table.qza \
--m-metadata-file metadata.tsv \
--m-metadata-column Group((any group which we want to do) \
--p-optimize-feature-selection \
--output-dir disease-ml

Top 10 genus ➖

```
qiime taxa collapse \
--i-table bacteria-asv-table.qza \
--i-taxonomy taxonomy1.qza \
--p-level 6 \
--o-collapsed-table genus-table.qza

qiime feature-table relative-frequency \
--i-table genus-table.qza \
--o-relative-frequency-table genus-relative.qza

qiime tools export \
--input-path genus-relative.qza \
--output-path exported-genus

biom convert \
-i exported-genus/feature-table.biom \
-o genus-relative.tsv \
--to-tsv
```

–Then in R:-

```
# ====================================
# Load library
# ====================================
library(tidyverse)

# ====================================
# Read QIIME TSV correctly
# ====================================
df <- read.delim(
"genus-relative.tsv",
skip = 1,
header = TRUE,
check.names = FALSE
)

# Set taxonomy names as row names
rownames(df) <- df[,1]
```

```
# Remove taxonomy column
df <- df[,-1]

# Check dimensions
dim(df)

# ====================================
# Find Top 10 genera
# ====================================
top10 <- sort(rowMeans(df), decreasing = TRUE)[1:10]
top10_names <- names(top10)

# Filter only top 10
df_top10 <- df[rownames(df) %in% top10_names, ]

# ====================================
# Convert to long format
# ====================================
df_long <- df_top10 %>%
as.data.frame() %>%
rownames_to_column("Genus") %>%
pivot_longer(
-Genus,
names_to = "Sample",
values_to = "Abundance"
)

# ====================================
# Plot stacked barplot
# ====================================
ggplot(df_long, aes(x = Sample, y = Abundance, fill = Genus)) +
geom_bar(stat = "identity", position = "fill") +
theme_classic() +
theme(
axis.text.x = element_text(angle = 90, hjust = 1),
text = element_text(size = 12)
) +
labs(
x = "Samples",
y = "Relative Abundance",
title = "Top 10 Genus Relative Abundance"
)
```

Finding genes and species of any biomarker or if not in taxonomy:-

In this first we go to taxa bar plots.qzv to see the desired bacterias abundance sample and then we should have ancom report showing the correct taxonomy and we need to find the exact feature ids with taxonomy.tsv and feature_table.tsv then we go to taxonomy.qvz and search the exact feature id and we also go to rep-seq.qzv and from there we select the sequence of the exact feature id and then we go to ncbi and then we know the organisms in that particular id.